Question			Answer	Marks	AO	Guidance	
8			Sketch diagram consistent with information in the question	**B1**	**2.5**	Triangle ADC or ABC or quadrilateral ABCD and 8 and 12 indicated eg side lengths or radii. Circles may or may not be shown.	
			$60 = 2 \times \frac{1}{2} \times 8 \times 12 \times \sin B$	**M1**	**3.1a**	**M1** implies previous **B1** These next 3 marks can be for angles B or D.	
			$\sin B = \frac{5}{8}$ so $B = 38.7°$ (0.675 rads)	**A1**	**1.1a**	One value of B or cos B	$\cos B = \frac{\sqrt{39}}{8}$
			OR $B = 141.3°$ (2.47 rads)	**A1**	**3.2a**	Other value of B or cos B	$\cos B = -\frac{\sqrt{39}}{8}$
			$AC^2 = 8^2 + 12^2 - 2 \times 8 \times 12 \cos 38.7$ $= 58.1$	**M1**	**3.1a**	Use of cosine rule	
			AC = 7.62 cm	**A1**	**1.1**	Accept 7.6 www	
			$AC^2 = 8^2 + 12^2 - 2 \times 8 \times 12 \cos 141.3$ $= 357.9$				
			AC = 18.9 cm	**A1**	**1.1**	Accept 19 www **A0** if more than 2 answers	
				[7]			